DEVIXO SOLUTIONS

AI/ML Internship Program

TASK 01

Python Programming, Data Analysis & Visualization

Pakistan House Prices Analysis (2023)

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Internship Domain	AI/ML
Submission Date	October 25, 2026
GitHub	https://github.com/shanaya3016/Task_01_AI_ML

1. Introduction

This report documents Task 01 of the Devixo Solutions AI/ML Internship Program. The task covers a complete data analysis workflow in Python: loading a real-world dataset, understanding its structure, cleaning and preprocessing it, performing exploratory and statistical analysis, creating visualizations and extracting insights. An optional interactive Streamlit dashboard was also built as a bonus.

2. Objective

The objective is to build practical skills in Python, Pandas, NumPy, Matplotlib and Seaborn, and to show the ability to clean data, analyse it statistically, visualise it and explain the findings clearly.

3. Dataset Name

Pakistan House Prices - 2023 (file: **house_prices.csv**)

4. Dataset Source

Kaggle: https://www.kaggle.com/datasets/manjitbaishya001/house-prices-2023

5. Dataset Description

The dataset contains property listings from major Pakistani cities, including the property type, location, city, price, number of bedrooms and bathrooms, area in Marla and the purpose of the listing (For Sale or For Rent). It mixes numerical and categorical variables, which makes it suitable for statistical analysis and visualization.

6. Dataset Size

Stage	Rows	Columns
Raw dataset	99,499	9
After removing the index column and duplicates	61,641	8
Final cleaned dataset (For Sale only)	43,340	8

7. Features / Columns Description

Column	Type	Description
property_type	Categorical	House, Flat, Upper Portion, Lower Portion, Penthouse, Room, Farm House
price_pkr	Numerical	Listing price in Pakistani Rupees (renamed from price)
location	Categorical	Area or society of the property (e.g. DHA Defence)
city	Categorical	Karachi, Lahore, Islamabad, Rawalpindi, Faisalabad
bathrooms	Numerical	Number of bathrooms (renamed from baths)
purpose	Categorical	For Sale or For Rent
bedrooms	Numerical	Number of bedrooms

Column	Type	Description
area_marla	Numerical	Property area in Marla (renamed from Area_in_Marla)

The dataset has no date or time column. The unnamed index column (Unnamed: 0) was dropped.

8. Methodology

The work followed the workflow given in the task document:

- Dataset selection and loading
- Data understanding (head, tail, sample, info, shape, column types)
- Data cleaning and preprocessing
- Exploratory data analysis and statistical analysis
- Data visualization with interpretation of every chart
- Insights, conclusion, documentation and GitHub organization

Tools used: Python 3, Google Colab, Pandas, NumPy, Matplotlib, Seaborn and Streamlit.

9. Data Loading

The libraries were imported and the CSV file was loaded into a Pandas DataFrame using *pd.read_csv()*. The first 5 rows, last 5 rows and a random sample of 5 rows were displayed, followed by df.info(), the column names, data types, missing values and the dataset shape (99,499 rows and 9 columns).

Column types: numerical columns were price, bathrooms, bedrooms and area. Categorical columns were property_type, location, city and purpose. No date/time columns were found. Identifying column types is important because numerical columns are used for statistics and correlation, while categorical columns are used for grouping and counting.

10. Data Cleaning

Step	Finding	Action
Missing values	0 in every column	No imputation needed
Index column	Unnamed: 0 carried no information	Dropped
Duplicates	37,858 duplicate rows (about 38%)	Removed and verified (0 remaining)
Column names	Inconsistent names	Renamed to price_pkr, bathrooms, area_marla
Data types	Text columns stored as object	property_type, city, purpose converted to category
Area <= 0	10 rows	Removed (zero area is impossible)
Bedrooms = 0	283 rows	Removed (not comparable)
Area > 100 Marla	41 rows	Removed as extreme outliers
Rent vs Sale	Prices on completely different scales	Rent rows (17,967) set aside; analysis on For Sale only

After cleaning, the dataset was verified: 0 missing values, 0 duplicates, correct data types and a final shape of 43,340 rows and 8 columns.

11. Exploratory Data Analysis

- **Most common property type:** House (29,041 listings).
- **Most common city:** Karachi (15,955 listings), followed by Lahore (14,483).
- **Most common location:** DHA Defence (2,528 listings).
- **Price range:** from about PKR 16,000 to PKR 44.9 million.
- **Outliers:** many high-price outliers exist in every city and a few very large plots (60 to 100 Marla).
- **Related variables:** price is most related to area, then bathrooms and bedrooms.

City	Mean price (PKR)	Median price (PKR)	Listings
Islamabad	14,635,707	12,500,000	6,598
Lahore	14,608,297	12,000,000	14,483
Karachi	12,893,316	9,500,000	15,955
Rawalpindi	12,114,902	9,500,000	4,779
Faisalabad	10,678,076	8,100,000	1,525

12. Statistical Analysis

Measure	Value	Meaning
Mean price	PKR 13,567,891	Average price, pulled up by luxury listings
Median price	PKR 10,900,000	Typical price, not affected by outliers
Std deviation	PKR 9,852,649	Prices vary widely around the mean
Skewness	1.12	Right-skewed distribution
Mode (type / city / location)	House / Karachi / DHA Defence	Most frequent categories

Mean vs median: the mean (13.57 million) is higher than the median (10.9 million). Because a few very expensive properties pull the mean upward, the median is the better measure of a typical price.

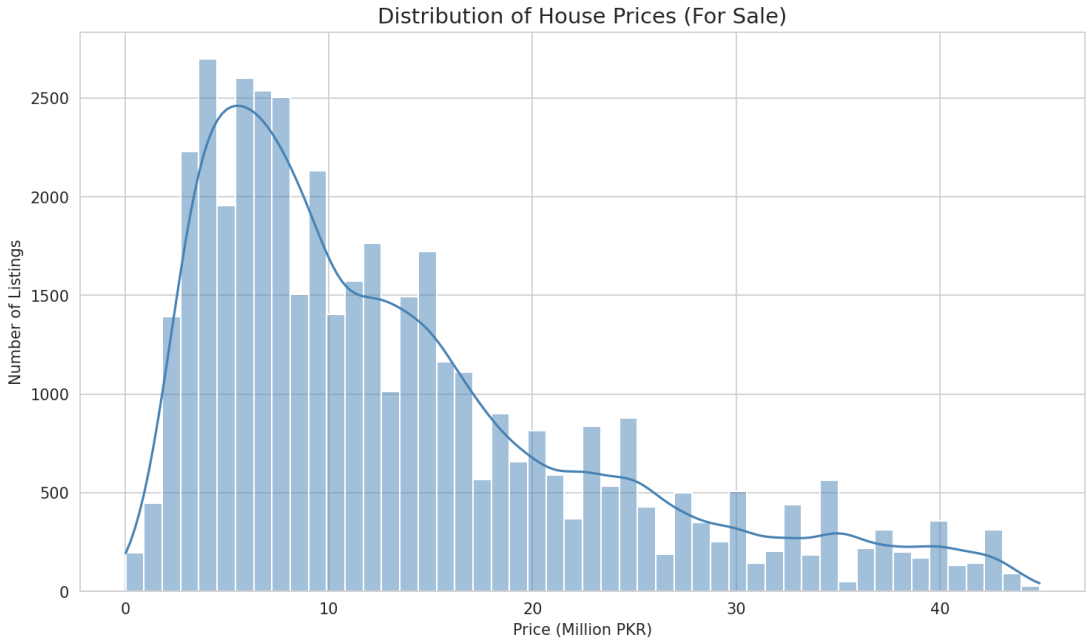
Correlation analysis:

Pair	Correlation	Strength
price and area	0.61	Strongest with price
price and bathrooms	0.58	Moderate
price and bedrooms	0.53	Moderate
bathrooms and bedrooms	0.83	Very strong
bathrooms and area	0.42	Moderate
bedrooms and area	0.38	Weak to moderate

All correlations are positive and none (except with itself) is close to 1, so price depends on several factors together, including city and location.

13. Data Visualizations

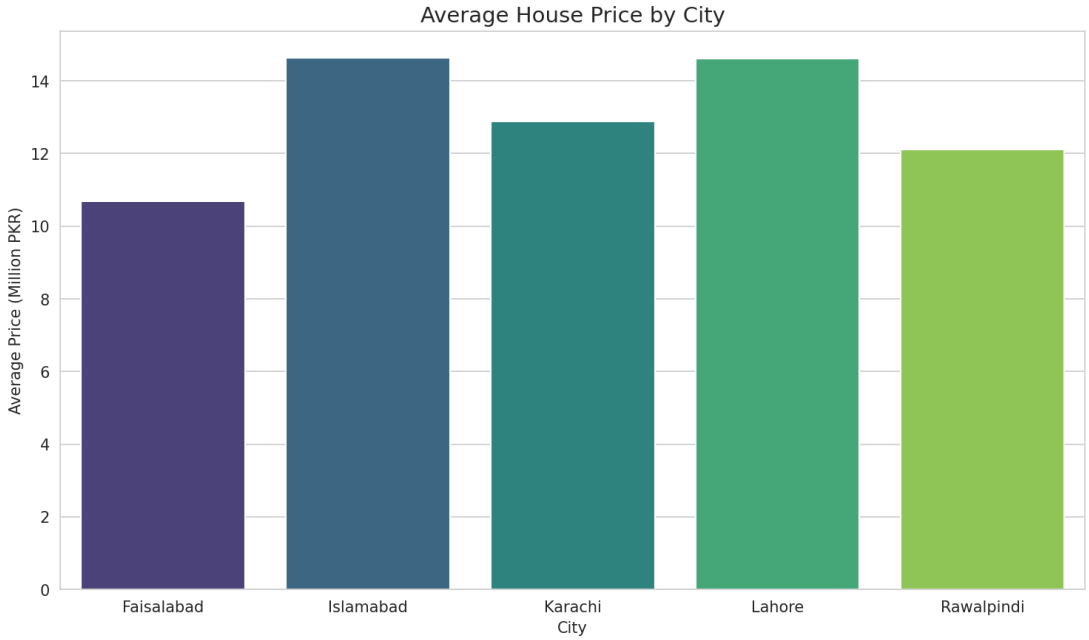
Visualization 1: Histogram of Price



Visualization 1: Histogram of Price

Interpretation: Most listings are priced between about 3 and 10 million PKR, with a long tail up to about 45 million. The distribution is right-skewed (1.12), which explains why the mean is higher than the median.

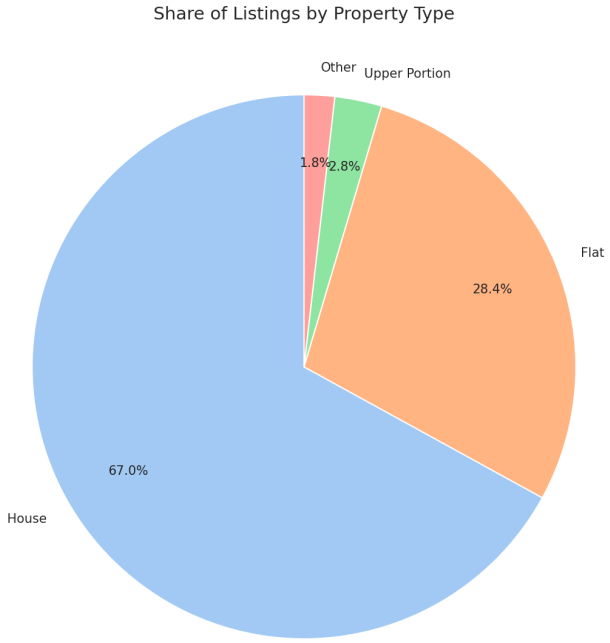
Visualization 2: Bar Chart - Average Price by City



Visualization 2: Bar Chart - Average Price by City

Interpretation: Islamabad and Lahore (about 14.6 million each) are the most expensive, followed by Karachi and Rawalpindi. Faisalabad is the cheapest (10.7 million). City is an important pricing factor.

Visualization 3: Pie Chart - Property Type Share



Visualization 3: Pie Chart - Property Type Share

Interpretation: Houses make up about two thirds of listings (67%) and Flats about 28%. All other types together are a small share, so averages mostly describe houses and flats.

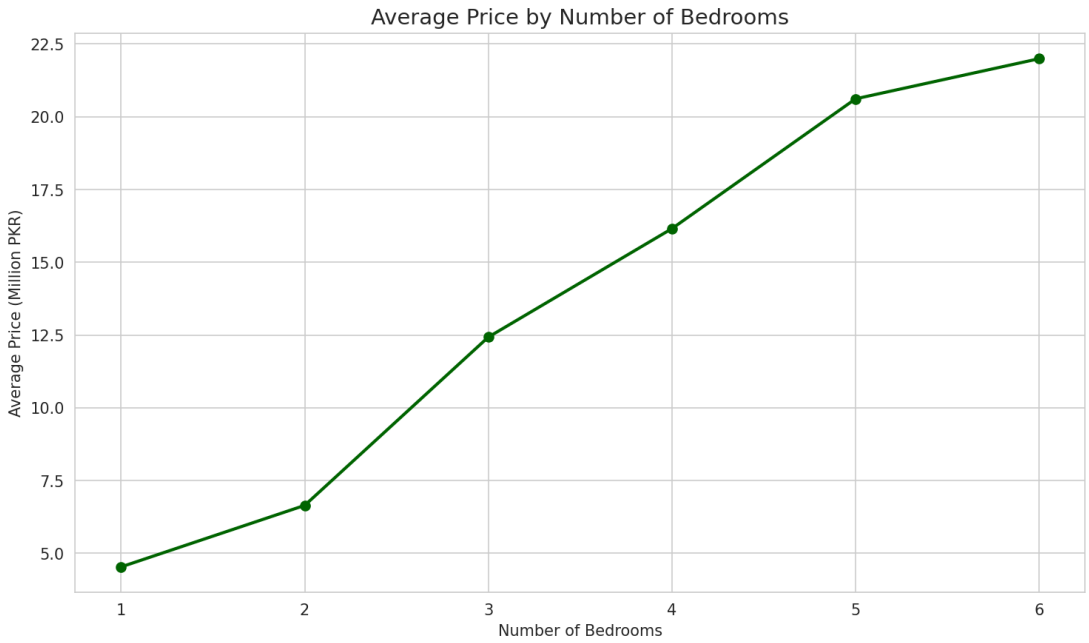
Visualization 4: Scatter Plot - Area vs Price



Visualization 4: Scatter Plot - Area vs Price

Interpretation: Within the main range (below 20 Marla) price rises with area. A vertical line at exactly 20 Marla (1 Kanal) shows a very common plot size. Above 40 Marla there are few listings and the relationship is scattered, so area alone cannot predict price.

Visualization 5: Line Chart - Average Price by Bedrooms



Visualization 5: Line Chart - Average Price by Bedrooms

Interpretation: The dataset has no date column, so the trend is shown over bedrooms. Average price rises steadily from about 4.5 million (1 bedroom) to about 22 million (6 bedrooms), with a slightly gentler slope after 5.

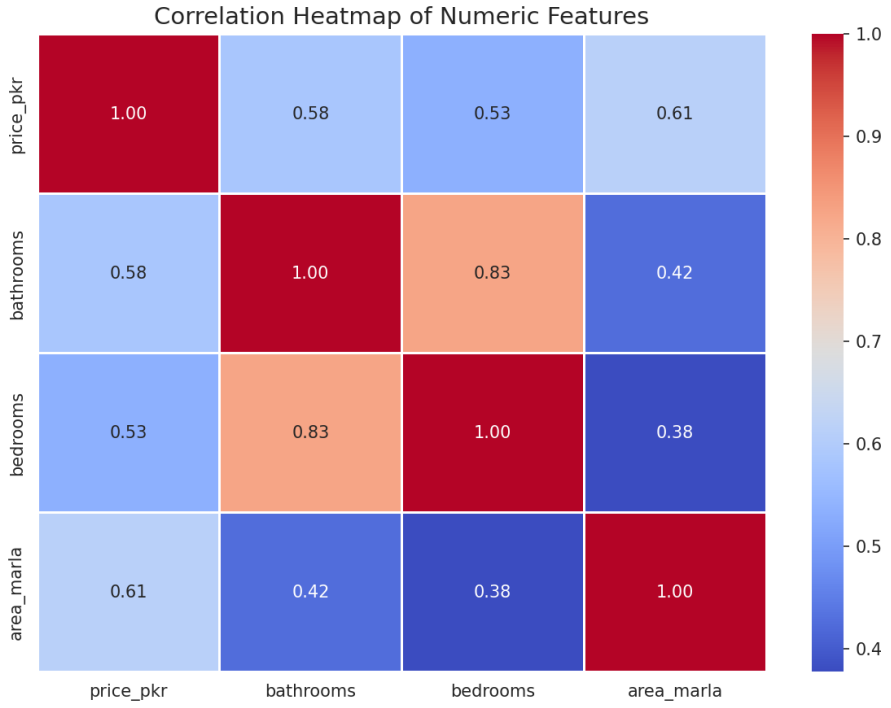
Visualization 6: Box Plot - Price by City



Visualization 6: Box Plot - Price by City

Interpretation: Islamabad and Lahore have the highest medians and Faisalabad the lowest. Every city has many high-price outliers, so the median is more reliable than the mean for comparing cities.

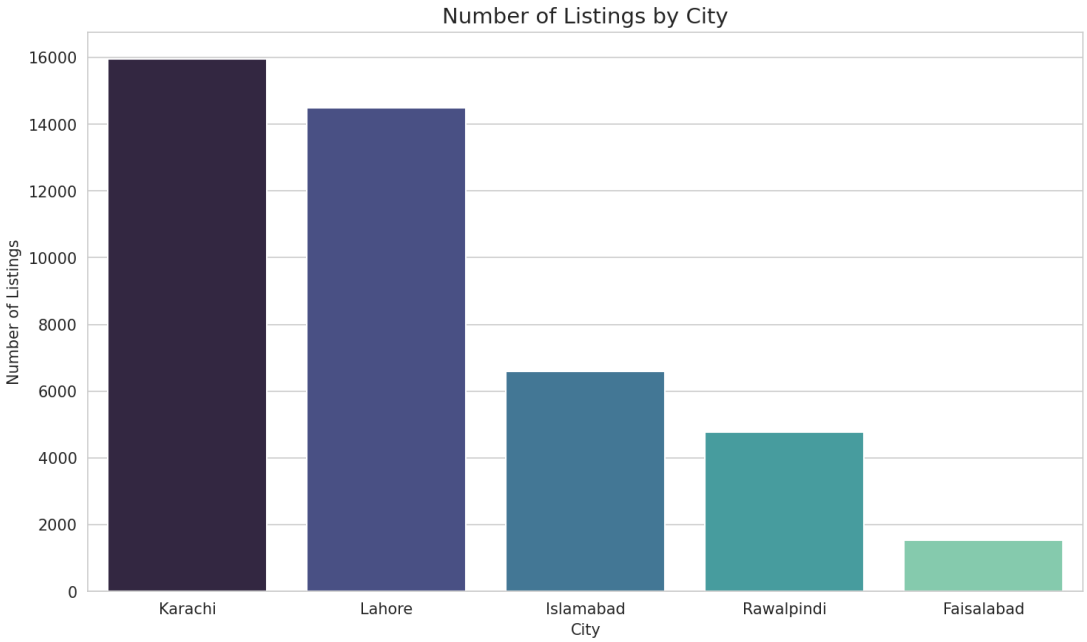
Visualization 7: Correlation Heatmap



Visualization 7: Correlation Heatmap

Interpretation: Bathrooms and bedrooms are the most strongly related (0.83). Price is most related to area (0.61). All values are positive. The overlap between bathrooms and bedrooms can cause multicollinearity in future ML models.

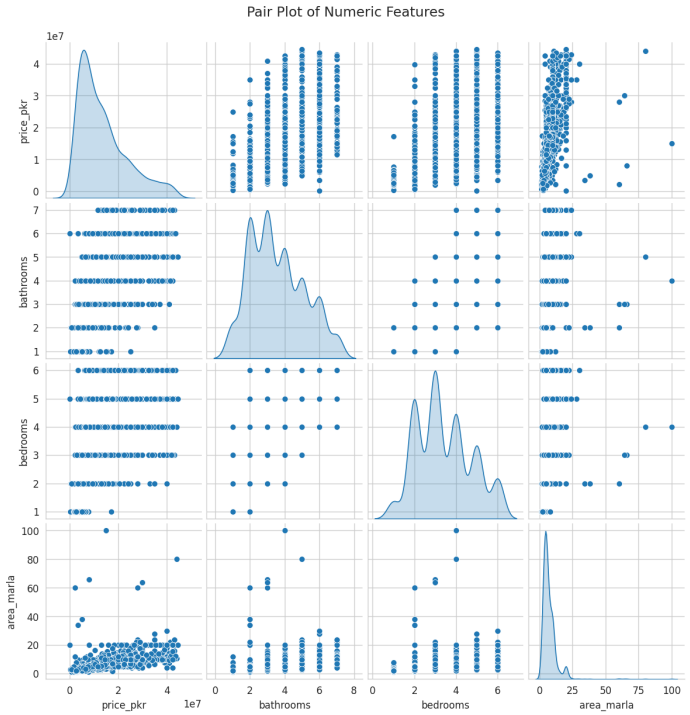
Visualization 8: Count Plot - Listings by City



Visualization 8: Count Plot - Listings by City

Interpretation: Karachi (about 16,000) and Lahore (about 14,500) together make up about 70% of the data. Faisalabad has few listings (about 1,500), so its average is less reliable.

Visualization 9: Pair Plot



Visualization 9: Pair Plot

Interpretation: Price and area are right-skewed. Bedrooms and bathrooms take whole numbers, forming lines of dots. A few very large plots stand apart. Skewed variables may need a log transform

before machine learning.

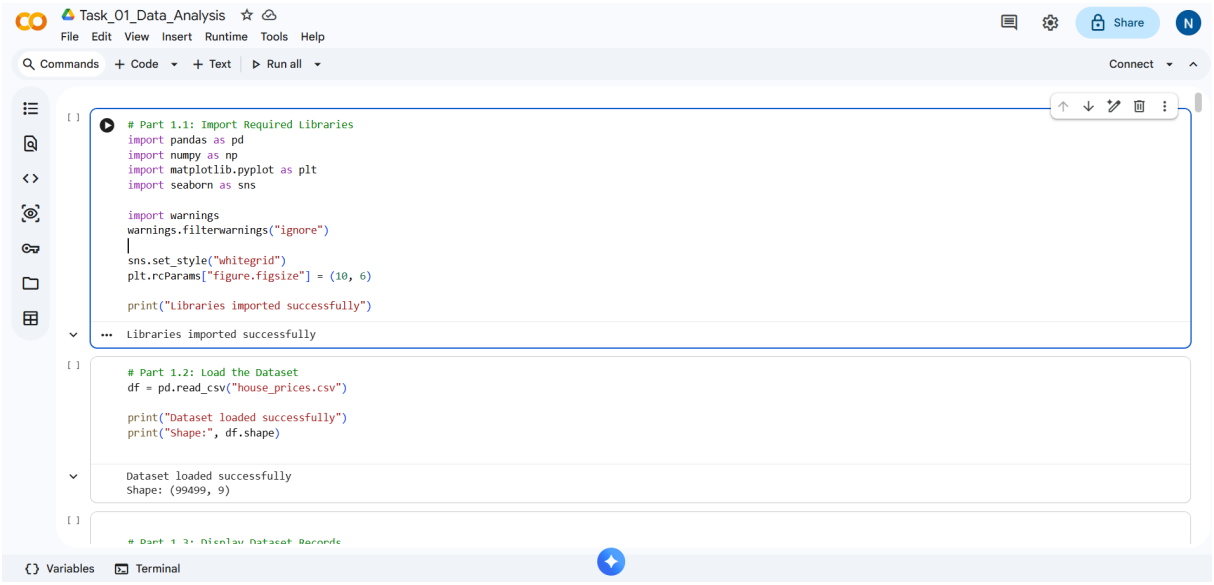
Visualization 10: Violin Plot - Price by Property Type



Visualization 10: Violin Plot - Price by Property Type

Interpretation: Houses have the highest median (about 12 to 13 million) and the widest spread. Flats have the lowest median (about 7 million). All types have long upper tails, so property type is an important price factor.

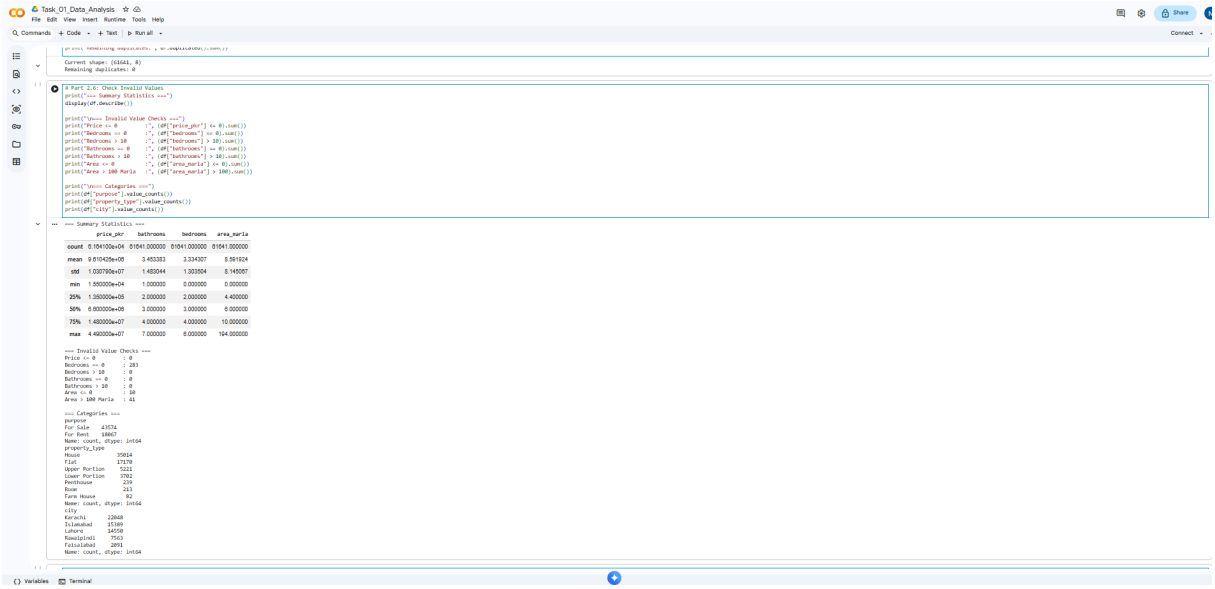
14. Code Screenshots



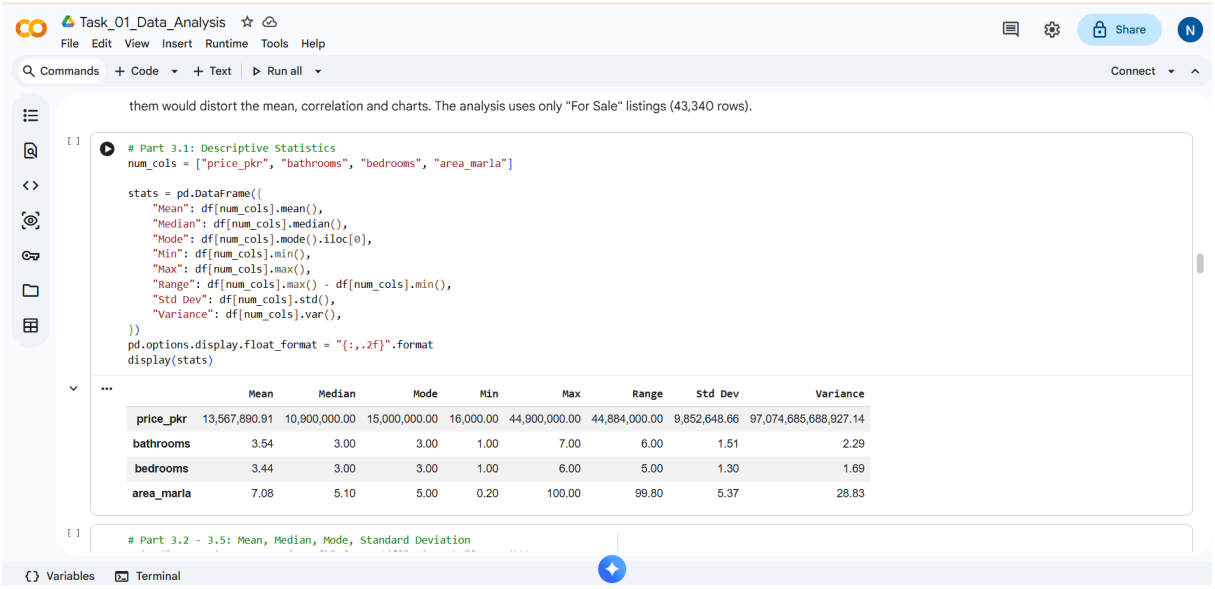
Libraries import and dataset loading



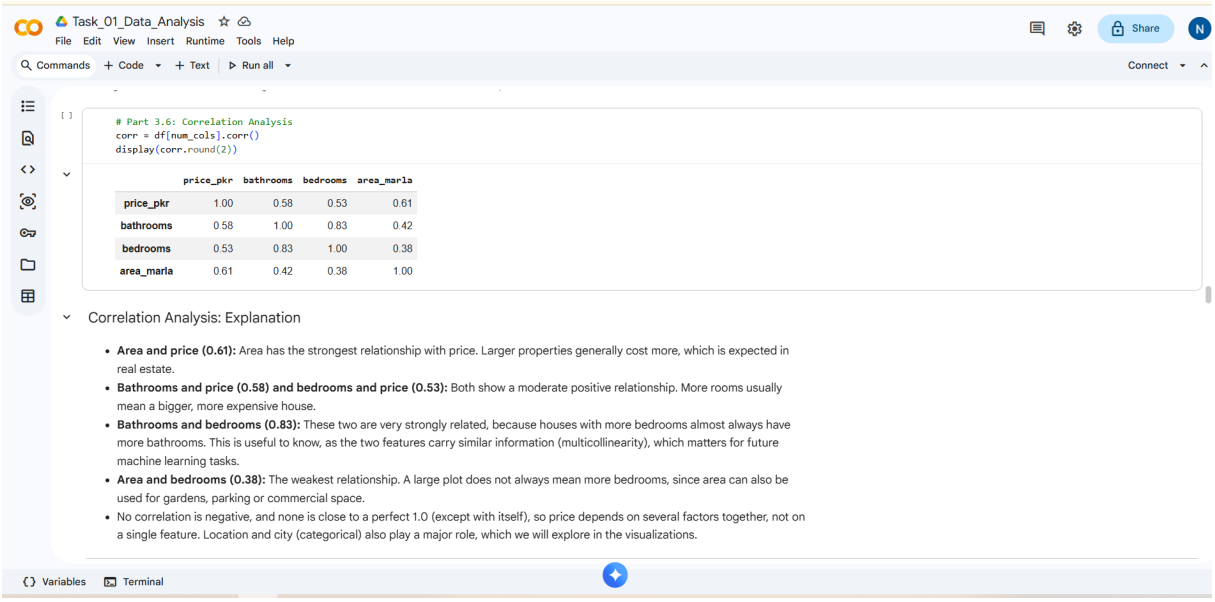
Data cleaning: duplicate records



Data cleaning: invalid values

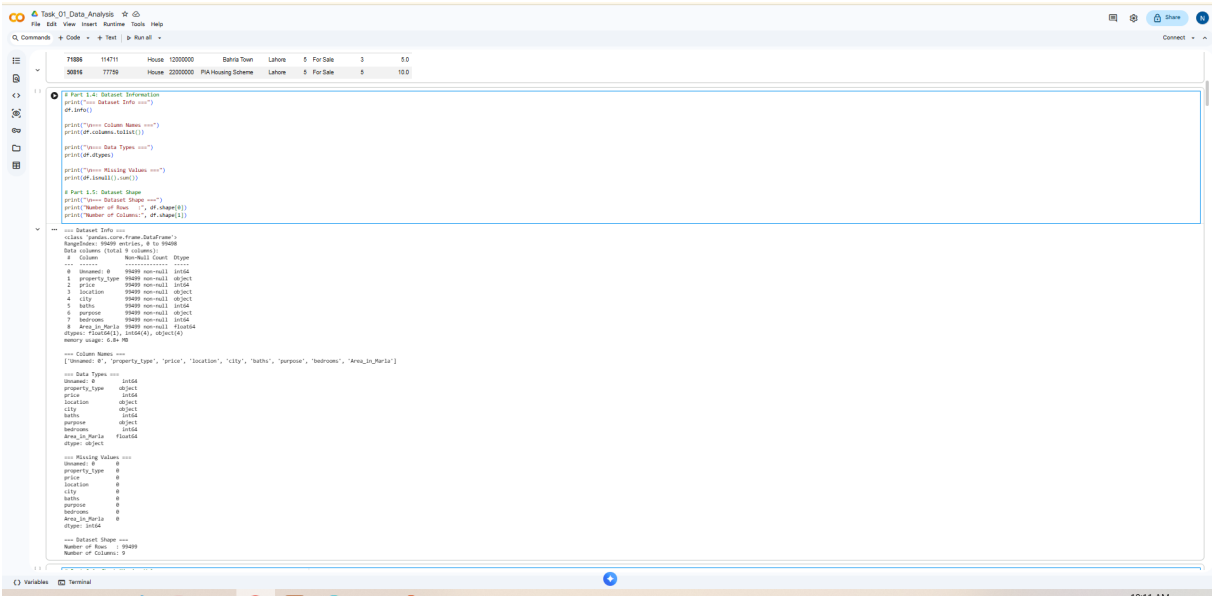


Descriptive Statistical analysis code



Correlation analysis code

15. Output Screenshots

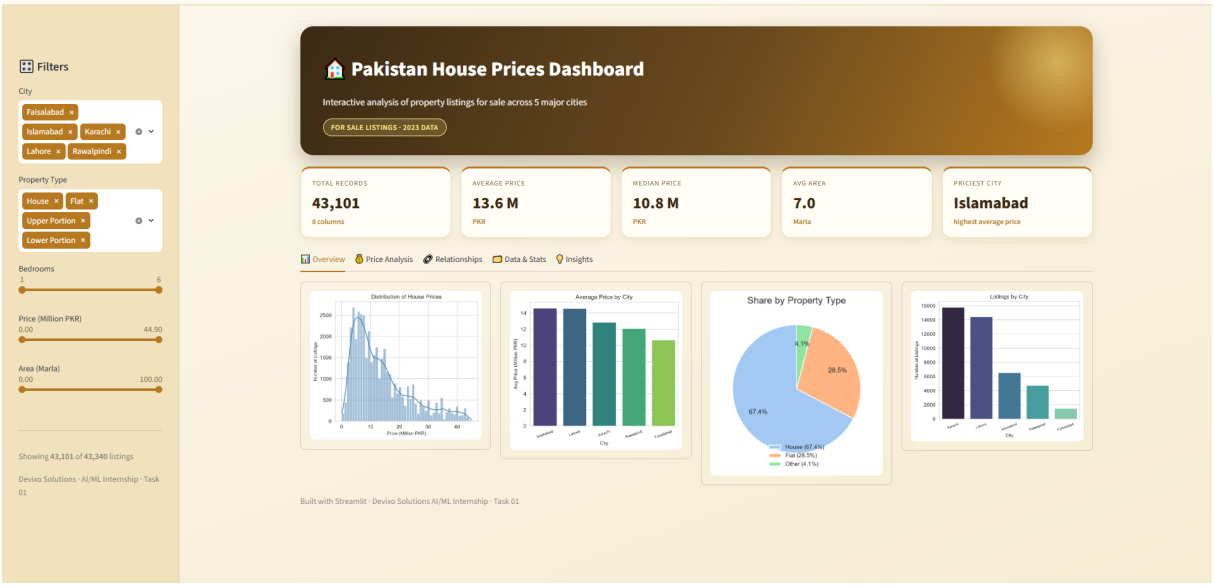


Dataset information, shape and missing values

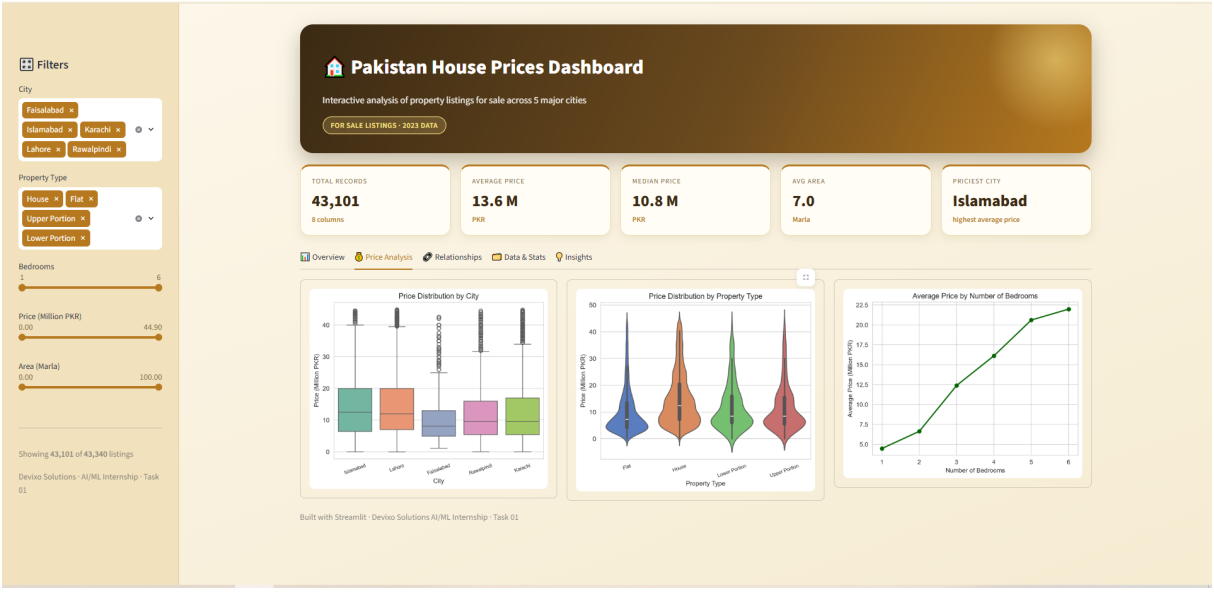
The cleaned dataset was verified with 0 missing values, 0 duplicates and a shape of (43340, 8).

16. Graph Screenshots and Streamlit Dashboard

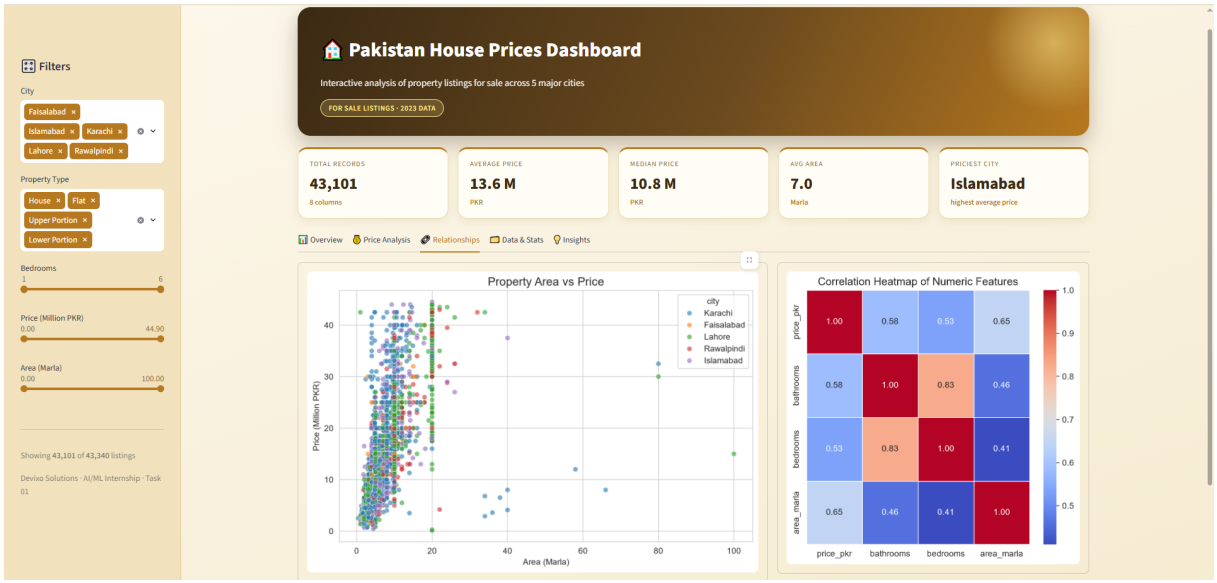
All 10 graphs are shown in Section 13. The optional Streamlit dashboard uses the same charts with interactive filters (city, property type, bedrooms, price and area), KPI cards, summary statistics, a correlation heatmap, key insights and a CSV download button.



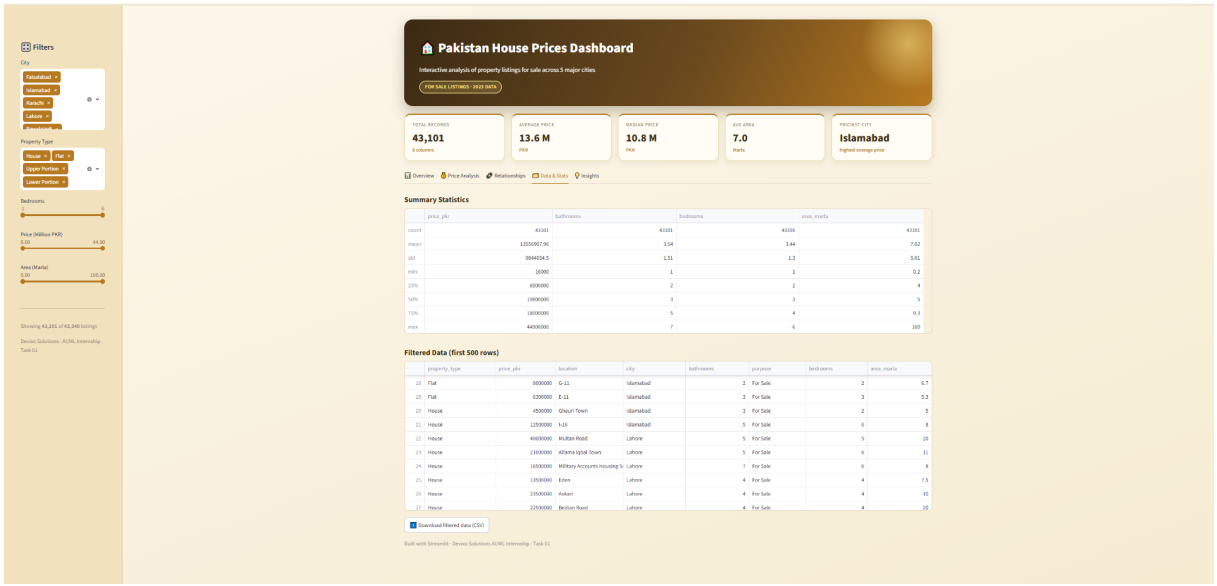
Dashboard: Overview tab



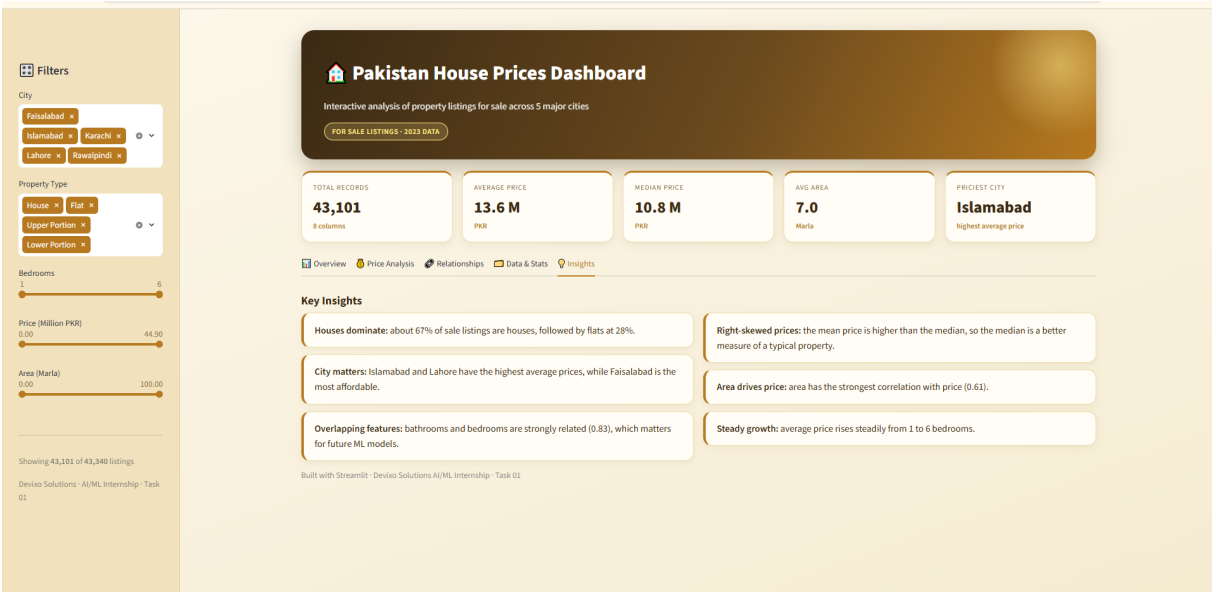
Dashboard: Price Analysis tab



Dashboard: Relationships tab



Dashboard: Data and Stats tab



Dashboard: Insights tab

17. Major Findings

- The raw data had about 38% duplicate rows and mixed rent and sale prices, which would have distorted the analysis.
- Sale prices are right-skewed: mean 13.57 million versus median 10.9 million PKR.
- Islamabad and Lahore are the most expensive cities; Faisalabad is the most affordable.
- Area has the strongest correlation with price (0.61); bathrooms and bedrooms are strongly related (0.83).
- Houses are 67% of listings and have a higher median price than flats.

18. Insights

1. **Houses dominate the market:** houses are 67% of sale listings (29,041), followed by flats at 28%.

2. **Prices are right-skewed:** mean 13.57M versus median 10.9M PKR (skewness 1.12) because a few luxury properties pull the average up.

3. **Islamabad and Lahore are the most expensive cities:** both average about 14.6M PKR, while Faisalabad is the cheapest at about 10.7M.

4. **Karachi has the most listings:** 15,955 (about 37%), with Lahore second at 14,483.

5. **Area is the strongest numeric driver of price:** correlation 0.61.

6. **Bathrooms and bedrooms are strongly related:** correlation 0.83, the highest in the dataset, which suggests overlapping information.

7. **Price rises steadily with bedrooms:** from about 4.5M PKR for 1 bedroom to about 22M PKR for 6 bedrooms.

8. **Prices vary widely:** the standard deviation (9.85M PKR) is large compared with the mean.

9. **Outliers exist in every city:** the box plot shows many high-price outliers, so the median is more reliable than the mean.

10. **20 Marla (1 Kanal) is a very common plot size:** the scatter plot shows a vertical line of listings at exactly 20 Marla.

11. **Data quality problems were found:** about 38% of raw records (37,858) were duplicates and rent and sale prices were mixed.

12. **Houses are priced higher than flats:** the violin plot shows a higher median price for houses.

19. Conclusion

In this project I analysed the Pakistan House Prices 2023 dataset from Kaggle. The raw data had 99,499 rows. During cleaning I removed the unnamed index column, 37,858 duplicate records and rows with invalid values (zero area, zero bedrooms and extreme areas above 100 Marla). I also renamed columns, corrected data types and kept only For Sale listings because rent and sale prices are on completely different scales. The final dataset has 43,340 rows and 8 columns. I then performed descriptive statistics, correlation analysis and created 10 visualizations, along with an interactive Streamlit dashboard.

The most important findings are that prices are right-skewed (mean 13.57M versus median 10.9M PKR), Islamabad and Lahore are the most expensive cities, and area has the strongest relationship with price (0.61). Houses make up two thirds of listings and cost more than flats. From this project I learned the complete data analysis workflow: cleaning real data, choosing suitable statistics, creating clear charts and interpreting them. The analysis can help buyers compare cities and property types, and it prepares the ground for future machine learning tasks such as house price prediction.

20. GitHub Repository Link

https://github.com/shanaya3016/Task_01_AI_ML